#3 · 10.7% of marks

Ch 6 — Portfolio Management

Tested every attempt: single-index risk split, Sharpe optimal portfolio & cut-off, CAPM / SML, APT, performance ratios, alternative investments (SGB).

1 Objectives & phases (SM)

Objectives (7)

Security / safety of principal · stability of income · capital growth · marketability · liquidity · diversification · **favourable tax status**

Phases (5)

Security analysis (risk-return of individual securities & correlation) → ***portfolio analysis*** (combine into feasible portfolios) → ***selection*** (efficient → optimal) → ***revision*** (monitor so it stays optimal) → ***evaluation*** (return & risk over the period)

Efficient portfolio: highest return among feasible portfolios with same / lower risk, or lowest risk with same / higher return.

2 Theories (SM)

- → ***Traditional approach***: concerned with the investor — study of age, health, responsibilities, needs for income / capital / liquidity, attitude to risk, tax status; objectives; investment strategy (fixed interest vs equity, high payout vs growth, tax, transaction cost vs switching gains, liquidity); diversification; selection of individual investments (intrinsic value)
- → ***Modern approach (Markowitz)***: risk of an individual asset hardly matters — what matters is its ***contribution to overall risk***; diversify with low / negatively correlated securities; higher return needs higher risk

3 Types of risk (SM)

	Systematic — external, macro, uncontrollable	Unsystematic — internal, micro, controllable
Types	***Interest rate*** (prices move inversely to rates) · ***Purchasing power / inflation*** (worst for bonds & fixed income) · ***Market*** risk	***Business*** risk (variability of operating profit) · ***Financial*** risk (capital structure / leverage, D/E)
Measure	Beta — by ***correlation method*** ($\beta = r_{im}\sigma_i/\sigma_m$) or ***regression method*** ($Y = \alpha + \beta X$)	Residual variance

Diversion of risk: as number of securities ↑ total risk ↓; ultimately only systematic risk remains.

4 Return & risk formulas

$E(R) = \Sigma\, p_i X_i$ · $\sigma^2 = \Sigma\, p_i(X_i - \bar{X})^2$ · $Cov_{AB} = r_{AB}\sigma_A\sigma_B$ · $r_{AB} = Cov_{AB} \div \sigma_A\sigma_B$

Two securities: $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2r_{12}\sigma_1\sigma_2$ · ***n securities***: $\sigma_p^2 = \Sigma\Sigma\, x_ix_j\sigma_{ij}$ (variance-covariance matrix)

SM cases: **r = +1** → $\sigma_p = x_1\sigma_1 + x_2\sigma_2$ (no diversification) · **r = −1** → $\sigma_p = |x_1\sigma_1 - x_2\sigma_2|$ (zero-risk mix possible) · **r = 0** → $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2$

App-computed: w 60 / 40, σ 20% / 30%, ρ 0.3 → σ_p = 19.35%; min-variance $w_A = (\sigma_B^2 - Cov) \div (\sigma_A^2 + \sigma_B^2 - 2Cov)$ = 76.6%.

5 Markowitz model (SM)

→ **Assumptions (SM 5.1)**: (i) return adequately summarises outcome · (ii) investors visualise a probability distribution of returns · (iii) risk estimates proportional to variance · (iv) decisions on two criteria — expected return & variance · (v) rational & risk-averse (prefer less risk for equal return) · (vi) yield in different scenarios used as return measure

→ **Efficient frontier** (Mean-Variance Dominance): a portfolio is **not efficient** if another has (a) higher return & lower σ, (b) higher return & same σ, or (c) same return & lower σ. Investor's choice on the frontier depends on risk-return preference

6 Capital market theory — CML

→ Tangent from R_f to market portfolio B → **CML** (R_f B Z). R_f–B = **lending portfolios** (part in risk-free); B–Z = **borrowing portfolios**

→ Slope = $(R_M - R_f) \div \sigma_M$ = reward per unit of risk · $\mathbf{E(R) = R_f + [(R_M - R_f) \div \sigma_M] \times \sigma_P}$

→ CML uses total risk σ (efficient portfolios); SML (CAPM) uses β (all securities)

7 Single index (Sharpe) model

Co-movement between stocks is due to the market index: $\mathbf{R_i = \alpha_i + \beta_i R_m + \in_i}$

$\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma_{\in i}^2$ · systematic risk = β_i^2 × variance of index · unsystematic = total − systematic · $\mathbf{Cov_{ij} = \beta_i\beta_j\sigma_m^2}$

Portfolio: $\sigma_p^2 = (\Sigma x_i\beta_i)^2\sigma_m^2 + \Sigma x_i^2\sigma_{\in i}^2$ · $R_p = \Sigma x_i(\alpha_i + \beta_i R_m)$ · α_p & β_p = weighted averages

App-computed: β 1.2, σ_m 15%, σ 25% → systematic 0.0324, unsystematic 0.0301. **MTP May 2026** (reverse): σ_p 25.03 → variance 626.50 = $1.73X^2 + 67$ → σ_m = **17.98**.

> ⚠ **TRAP**
> Unsystematic portfolio risk uses **weights squared** ($\Sigma x^2\sigma_{\in}^2$). Keep variance units consistent (15% → 225 or 0.0225).

8 APT (SM)

→ Alternative to CAPM; uses risk premia of several macro factors: $\mathbf{E(R_i) = R_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \ldots + \lambda_n\beta_n}$

→ No surprises → actual = expected. With **unanticipated changes**: $E(R) = R_f + \beta_1(EV_1 - AV_1) + \ldots$ (surprise factor = expected − actual value of factor)

App-computed: R_f 6%, premia 4% & 2%, sensitivities 1.2 & 0.8 → **12.4%**. May 2026: target sensitivities via long / short amounts — two simultaneous equations.

8-MARK FAVOURITE

⑨ Steps (SM)

1. Find "**excess return to beta**" $(R_i - R_f) \div \beta_i$ — i.e. Treynor's ratio — for each stock
2. Rank highest to lowest
3. Compute C_i in ranked order: $C_i = \sigma_m^2 \Sigma[(R_i - R_f)\beta_i/\sigma_{ei}^2] \div [1 + \sigma_m^2 \Sigma(\beta_i^2/\sigma_{ei}^2)]$
4. C^* = highest C_i; select stocks whose excess-return-to-beta is above C^*, reject the rest
5. Weights: $Z_i = (\beta_i/\sigma_{ei}^2) \times [(R_i - R_f)/\beta_i - C^*]$; $X_i = Z_i \div \Sigma Z_i$

App-computed example: R_f 5%, σ_m^2 10.

Sec	R_i	β	σ_e^2	$(R-R_f)/\beta$	Rank	C_i
A	15	1.0	30	10.0	1	2.500
B	17	1.5	40	8.0	2	4.132
D	11	0.8	10	7.5	3	4.982
C	12	1.0	20	7.0	4	**5.314 = C^***
E	8	1.2	35	2.5	5	4.978 (falls → stop)

Sec	Z_i	Weight
A	0.1562	30.27%
B	0.1007	19.52%
D	0.1749	33.88%
C	0.0843	16.33%

⚠ TRAP

Ranking uses **excess return ÷ β**, not ÷ σ. If the question **gives** C^*, go straight to Z_i (May 2026 Q4(a)).

FORMULA CORE

10 CAPM (SM)

- → **$R_j = R_f + \beta(R_m - R_f)$**; risk premium of a stock = β × market risk premium. Graph = **SML** (required return vs β); market β = 1
- → SM: only **one risk-free rate**; if two are given, aggressive = higher, conservative = lower, moderate = average (Illus. 5)
- → **Decisions (SM 8.2)**: CAPM required < actual / expected return → **buy** (under-valued) · required > actual → **sell** (over-valued) · equal → **hold**. Price basis: actual price < CAPM price → under-valued
- → **Advantages**: (i) risk-adjusted return — usable as risk-adjusted discount rate; (ii) cost of equity for a **no-dividend company**
- → **Limitations**: (a) reliability of beta; (b) ignores other (unsystematic) risks; (c) information available — R_f and market return hard to estimate
- → Back-solving (Jan 2026 MCQ): two portfolios' returns & betas → solve R_f and R_m

App-computed: 6 + 1.3 × (14 − 6) = **16.4%**. SM Illus. 2: β 1.40, R_m 14, R_f 10 → 15.6%; premium +2.5 points → 19.1%.

11 Portfolio evaluation (SM)

Measure	Formula	SM meaning
Sharpe ratio	$(R_i - R_f) \div \sigma_i$	Risk premium per unit of **total** risk
Treynor ratio	$(R_i - R_f) \div \beta_i$	Risk premium per unit of **systematic** risk
Jensen's alpha	$R_p - [R_f + \beta(R_m - R_f)]$	Actual return minus CAPM-expected return

App-computed: R_p 15, σ 18, β 1.1, R_f 6, R_m 13 → Sharpe 0.50; Treynor 8.18; Jensen +1.30%.

12 Mixed portfolios (RTP Nov 2026)

Scenario returns × probabilities → E(R), σ, covariance → portfolio variance. Risk-averse → lowest σ for acceptable return; risk-seeker → highest return.

13 Portfolio strategy (SM)

→ **Active (APS) — 4 principles**: **market timing** (business cycle, moving average, advance-decline, econometric models) · **sector rotation** · **security selection** · **use of specialised investment concept**

→ **Passive — 2 guidelines**: (a) well-diversified portfolio at a predetermined risk level; (b) hold relatively unchanged unless inadequately diversified or inconsistent with preferences. Passively managed fund = **index fund** (doesn't remove market risk)

14 Selection of securities (SM)

Bonds — 4 factors

Yield to maturity · **tax shield** · **liquidity** · **risk of default** (credit rating / D/E, interest cover, earning power, industry)

Equity — 3 approaches

Technical (price & volume) · **fundamental** (intrinsic value vs price → buy / hold / sell) · **random selection** (efficient market, fairly priced)

15 Revision & rebalancing (SM)

	Buy & hold ("do nothing")	Constant mix	CPPI
Mechanism	No rebalancing; linear exposure to stocks	Rebalance to fixed % — buy stocks as they fall, sell as they rise	Floor in non-fluctuating assets; cushion in shares
Pay-off	Straight	**Concave**	**Convex**
Down / up market	Definitely poor in down	Not much in down, relatively poor in up	Good in down, well in up
Flat market	Between the other two	**Does well**	**Performs poorly**

16 Asset allocation strategies (SM – 4)

Integrated	Examines market conditions + investor objectives & constraints; allocation incorporating market forecast
Strategic	Optimal mix from historical returns, risk, covariances; periodically restored to target
Tactical	Risk tolerance constant; allocation changed on expected market conditions
Insured	Risk exposure adjusted with wealth — more value, more ability to take risk

17 Fixed income portfolio (SM)

- → **Process (5)**: set objective → investment policy guidelines → strategy (active / passive) → select securities → evaluate vs benchmark. **Return measures**: arithmetic average, time-weighted, rupee-weighted, annualised
- → **Passive**: **buy & hold** (may set duration = benchmark) · **indexation** (replicate bond index) · **immunization** (hybrid; popular with pension funds) · **matching cash flows** (zero-coupon bonds to meet promised payments)
- → **Active**: (1) **forecasting returns & interest rates** — **bullet** (all in one maturity), **barbell** (equal short & long), **ladder** (equal across maturities) · (2) **bond swaps** — **pure yield pickup**, **substitution**, **inter-market (international) spread**, **tax swap** · (3) **interest rate swap**

18 Alternative investments (SM)

- → **Features (6)**: high fees · limited historical data · illiquidity · less transparency · extensive research required · leveraged buying. **Types**: real estate, gold, private equity, REITs, hedge funds, ETFs, MFs, commodities, distressed securities
- → **Real estate** — valuation complex (inefficient market, illiquidity, only approximate comparison, high transaction cost…). **4 approaches**: sales comparison (like P/E), income (perpetual net income ÷ required return), cost (replacement cost + land value, adjusted for location), discounted after-tax cash flow (NPV-like)
- → **Gold forms**: jewellery (making charges) · **gold bars / coins** (storage cost) · **SGBs** — government securities in grams of gold, issued by **RBI on behalf of GoI**, paid & redeemed in cash, quantity of gold protected (market price at redemption), no storage risk / cost, **periodical interest**, held with RBI or in demat · **Gold ETFs** (price based on gold; full transparency; low expenses) · **E-gold** (NSEL, 2010; 1 unit = 1 gram in demat)
- → **Distressed securities**: securities of firms in / near bankruptcy, bought cheap to revive; arbitrage by long debt + short equity. **Risks**: liquidity, event, market, **human** (judge's decision)

SGB scheme specifics used in May 2026 MCQ (2.5% p.a. interest, 8-year tenor, redemption gain exempt for individuals) are **not stated in the SM** — use only the data given in the question.

19 Correlation — how Ch 6 actually connects

Ch 6 idea	The actual link (app-written cross-reference; the Ch 6 side is SM)
CAPM K_e	**Ch 5**: the K_e found here is the discount rate in DDM / FCFE; a stock's equity risk premium = $\beta \times (R_m - R_f)$, not $R_m - R_f$.
$\beta = r_{im}\sigma_i/\sigma_m = \text{Cov} \div \sigma_m^2$	**Ch 9**: β_p (weighted average) decides the number of index futures to sell / buy to change portfolio beta.
Sharpe, Treynor, Jensen	**Ch 8**: same formulas rank mutual funds. Sharpe uses σ (total risk — fund is the whole investment); Treynor uses β (well-diversified); Jensen $\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$.
Fixed-income strategies	**Ch 5**: immunization = duration equal to horizon; bullet / barbell / ladder are duration choices; IRS (Ch 12) changes the duration of the portfolio.
Active vs passive	**Ch 4**: if markets are efficient, passive (index) wins after costs; active needs mispricing. **Ch 8**: index funds / ETFs are the passive vehicle.
Portfolio σ	**Ch 2**: σ_p feeds VaR; lower correlation → lower σ_p → lower VaR.